Jiayi Lyu

M.S. Thesis Student · Electrical and Computer Engineering

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Objectives

To pursue a Ph.D. in Computer Science or Electrical and Computer Engineering beginning in Fall 2027, with a focus on efficient and reliable AI systems. Building on my research in vision-language models, environmental time-series prediction, agent AI, and edge AI, I aim to develop methods that connect model design with real-world data and computational constraints, while strengthening my ability to lead independent research.

Fields of Interest

• Efficient AI and Machine Learning Systems • Vision-Language Models and Multimodal Learning • Computer Vision
• Memory-Based Agents and Time-Series Prediction • Edge AI • Hardware-Software Co-Design

Education

Southern Illinois University Carbondale (R1) — *Carbondale, USA*
M.S. (Thesis) in Electrical and Computer Engineering — *Aug. 2025 – Present*

- Advisor: Dr. Chao Lu, Professor, School of Electrical, Computer and Biomedical Engineering
- Title: Leveraging Artificial Intelligence for Climate-Smart Soybean Farming in Sustainable Agriculture
- Expected graduation: May 2027
- GPA: 4.0

Hebei University — *Baoding, China*
B.S. in Data Science and Big Data Technology — *Sep. 2021 – June 2025*

- Undergraduate research advisor: Shuo Wang
- Undergraduate research: Social Recommendation Algorithms Based on Social Network Structure
- GPA: 3.9

Professional Experience

Aug. 2025 – Present

Graduate Research Assistant, School of Electrical, Computer and Biomedical Engineering, Southern Illinois University Carbondale.

- Developed image and environmental time-series datasets and machine learning models for agricultural vision, vision-language adaptation, and rare-event prediction.
- Evaluated models through baseline comparisons, ablation and quantization studies, and edge-deployment profiling; prepared research figures, tables, and manuscripts.

Aug. 2025 – May. 2026

Graduate Teaching Assistant, School of Computing, Southern Illinois University Carbondale.

- CS 330/491: Introduction to the Design and Analysis of Algorithms (Fall 2025))
- CS 434/526: Learning from Data (Spring 2026)
- **Completed teaching assistant training and passed the International Teaching Assistant (ITA) oral examination.**
- Supported lectures and office hours, answered student questions, graded assignments, maintained grade records, and proctored quizzes and examinations.

Jul. 2024 – Aug. 2024

Software Development Intern, Beijing Qingcheng Boya Education Technology Co., Ltd., Hebei, China.

- Contributed to requirements analysis and front-end and back-end development; integrated facial recognition for image uploads and entry/exit logging.
- Created and maintained databases and optimized data storage, queries, and updates.

Publications

Published Student Abstract

1. Y. Yan, M. W. Chen, **Jiayi Lyu**, C. Zhao, H. Gao, and Z. Chen, **"Fine-tuning Zero-shot Large Language Models for Patient-reported Outcomes (Student Abstract),"** AAAI Conference on Artificial Intelligence (AAAI 2026), Student Abstract Track, 2026.

Manuscripts Under Review

1. **Jiayi Lyu**, C. Zhao, Z. Wang, and Z. Chen, **"SEMA: A Structured Environmental Memory Agent for Time-Series Oyster-Associated Norovirus Outbreak Prediction,"** AAAI 2027. Under review.
2. **Jiayi Lyu**, F. Ravelombola, Z. Chen, F. Lin, and C. Lu, **"Learning from Limited Expert-Labeled Data: Vision-Language Models for Recognizing Purple Seed Stain Symptoms in Soybean Seeds,"** 18 pages. Information Processing in Agriculture. (Journal Impact Factor: 8.9) Under review.
3. R. Maharjan, **Jiayi Lyu**, F. Ravelombola, F. Lin, and C. Lu, **"EdgeWiltNet: SuperImage-Enhanced Soybean Wilting Severity Classification Using UAV Imagery,"** 17 pages. Information Processing in Agriculture. Under review.

Research Projects (Graduate Level)

- As a graduate research assistant, I collaborated with my advisor on AI-based soybean analysis, covering purple seed stain classification with limited labeled data and soybean wilting classification from UAV imagery. For purple seed stain, a single-seed dataset was developed using SAM2 segmentation, geometric filtering, quality control, and expert label verification. Vision-language approaches, including zero-shot CLIP, prompt variations, frozen-feature classifiers, adapters, and LoRA fine-tuning, were evaluated. For soybean wilting, SuperImage-based super-resolution was combined with a lightweight network incorporating depthwise-separable convolutions and squeeze-and-excitation attention. Model performance was assessed through baseline comparisons, ablation studies, and TensorFlow Lite quantization experiments, comparing classification accuracy, model size, and inference latency.

  Project title and funding source: **USDA-NIFA: Leveraging Artificial Intelligence for Climate-Smart Soybean Farming in Sustainable Agriculture: Research and Educational Initiatives, 7/1/24 – 6/30/27.**

- As a graduate research assistant, I collaborated with an SIU faculty member in computer science on oyster-associated norovirus outbreak prediction, contributing to dataset construction and manuscript preparation. Environmental records from USGS, NOAA, and NSRDB were integrated with harvest-area data, standardized, and cross-checked against outbreak labels. An interpretable agent-based framework was developed, using deterministic agents to encode lagged environmental features and maintain shared historical memory. A risk-reasoning agent combined current evidence with historical context, while a verification agent checked supporting evidence.
  Project title and funding source: **USDA-NIFA: DSFAS: AI-Driven Predictive Modeling for Early Detection of Norovirus Outbreaks in Oyster and Shellfish Aquaculture, 2/1/26 – 1/31/28.**

Additional research: I collaborated on LLM fine-tuning for patient-reported outcome classification and toxicity prediction in prostate radiotherapy.

Selected Coursework

• ECE 510: Hardware Designs and Architectures for AI • ECE 511: Software-Hardware Co-design for Deep Neural Networks
• ECE 513: Digital VLSI Design • ECE 516: Implementation of VLSI Systems with HDL • ECE 528: Programmable ASIC Design
• ECE 514: Design of Intelligent Embedded Systems

Course Projects

- **VLSI Implementation of Quantized Neural Networks** (Fall 2026, ECE 516): Implemented a three-layer neural network in Verilog with 8-bit quantized weights for MNIST digit classification.
- **Knowledge Distillation for Deep Neural Networks on Embedded Systems** (Fall 2026, ECE 514): Exploring teacher–student knowledge distillation to reduce model complexity while preserving accuracy on resource-constrained systems.
- **Dataset Difficulty with V-Usable Information** (Fall 2025, ECE 511): Compared BERT models trained on original and transformed SNLI datasets using pointwise V-information to analyze annotation artifacts and lexical shortcuts.

Professional Services

- Reviewer at **IEEE Sensors Journal**.

Presentations, Outreach and Service

- **Research Presentations — University of Missouri** *Nov. 14, 2025; Oct. 15, 2026*
  Presented research on soybean seed quality and discussed agricultural AI projects with collaborators.
- **Research Outreach — Du Quoin State Fair** *Aug. 30, 2025; Sep. 5, 2026*
  Explained research posters to visitors at the SIU outreach booth and invited visitors to complete a brief survey.
- **Team Leader — "Internet+" Innovation and Entrepreneurship Competition** *Mar. 2024*
  Coordinated the team's project development.
- **Campus Service — Hebei University**
  Volunteered during the COVID-19 pandemic. *Sep. 2022*
  Organized student union activities. *Nov. 2021*

Awards and Funding

- **Graduate Student Professional Development Funding** *2025*
  Southern Illinois University.
- **Outstanding Student Scholarship (Third Class)** *2024–2025*
  Hebei University.
- **Outstanding Student Scholarship (Second Class)** *2023–2024*
  Hebei University.
- **Outstanding Student Scholarship (Third Class)** *2022–2023*
  Hebei University.

Technical Skills and Test Scores

- **Tools:** Python (NumPy, pandas, Matplotlib, PyTorch); TensorFlow (TensorFlow Lite); Hugging Face (Transformers, PEFT); MATLAB; MySQL; Verilog (Vivado); Cadence Virtuoso; Edge Impulse; Linux; Git; LaTeX (Overleaf/TexStudio).
- **Languages:** Mandarin (native); English (TOEFL: 85).
- **Test:** GRE: 323 (V 155 + **Q 168** + AW 3.5).